Bhavya Paruchuri Data Analyst

bhavya@myworkmails.com | 945 (266)-0261 | USA | LinkedIn

Summary

Data Analyst with 3+ years of experience in data extraction, transformation, and analysis to support business decision-making. Expert in SQL, Python, and R for statistical analysis, data cleaning, and modeling. Skilled in developing ETL pipelines and working with cloud platforms like AWS and Azure. Experienced in creating interactive dashboards using Power BI and Tableau to visualize key metrics. Strong collaborator with cross-functional teams, adept at applying analytical techniques to uncover insights and improve data-driven strategies.

Technical Skills

- **Data Analysis & Statistical Tools**: Python (NumPy, pandas, scipy), R (logistic regression, AUC-ROC)
- **Databases & Querying**: SQL (window functions, CTEs), SQL Server
- **Cloud & ETL**: AWS (S3, Glue, Athena, Lambda), Azure (Data Factory, Synapse)
- **Data Visualization**: Power BI (DAX, KPIs, slicers), Tableau (Level of Detail calculations)
- **Techniques**: Exploratory Data Analysis (EDA), Cohort Analysis, Customer Segmentation (K-means, hierarchical clustering), Time-Series Decomposition, Data Validation and Cleaning
- **Methodologies**: Agile Sprint Collaboration, Requirement Gathering

Professional Experience

Data Analyst, Mizuho Financial Group 03/2024 – Present | Remote, USA

- Worked on customer segmentation analysis by collaborating with marketing, finance, and IT teams in Agile sprints and conducting requirement gathering sessions to identify customer segments and assess profitability for targeted marketing and product development strategies.
- Used SQL with window functions and CTEs for complex querying and database optimization. Processed large-scale transactional and demographic data stored in AWS S3 in CSV and Parquet formats using AWS Glue and Athena to build efficient ETL pipelines.
- Applied exploratory data analysis and cohort analysis to uncover customer behavior patterns and segmented customers using K-means and hierarchical clustering resulting in an 18% increase in marketing campaign response rates.
- Programmed in Python using NumPy, pandas, and scipy for statistical analysis and trend detection. Performed correlation analysis and time-series decomposition to understand segment profitability fluctuations.
- Executed data validation and cleaning processes using AWS Glue workflows and Lambda functions which reduced data errors by 25% and improved the reliability of downstream analysis.
- Developed interactive Power BI dashboards featuring drill-down filters, KPIs, and dynamic slicers. Implemented DAX calculations to display customer lifetime value and segment revenue trends which enhanced executive decision-making with real-time insights.

Data Analyst, The Cigna Group 01/2021 – 12/2022 | Andhra Pradesh, India

- Collected customer data for the Health Insurance Customer Retention Analysis project and collaborated with marketing, product, and data engineering teams in agile sprints to gather requirements and reduce customer churn by 18.6% over two years.
- Used SQL Server to query customer policy, claim, and engagement data and built ETL pipelines with Azure Data Factory and Azure Synapse to reduce data processing time by 43% and support predictive model inputs.
- Performed exploratory data analysis and cohort analysis to uncover lapse trends in customer behavior and identified age, claim frequency, and renewal delays as key churn predictors that reduced high-risk segment churn by 12.3%.
- Used Python with NumPy and Pandas for data cleaning, feature engineering, and segmentation of over millions of policy records to support churn modeling and customer lifecycle analysis across multiple policy product categories.
- Developed logistic regression models in R to predict customer churn and used AUC-ROC metrics to validate model accuracy at 0.87, identifying three top features that influenced churn across policyholder age and claim categories.
- Executed data validation steps within Azure Data Factory pipelines and applied ETL checks on structured and unstructured data which improved overall data integrity by 29% and reduced model training errors in production environments.
- Created Tableau dashboards to visualize churn KPIs with region and product filters and implemented Level of Detail calculations to display churn by policy type which improved executive decision-making accuracy by 35% quarterly.

Education

Master of Information Technology Management 01/2023 – 12/2024
Indiana Wesleyan University (IWU), Marion, Indiana, USA

Bachelor of Computer Science Engineering 08/2018 – 08/2022
Vignan's Lara Institute of Technology and Science, Guntur, Andhra Pradesh, India